

Nowcasting Rwanda's quarterly GDP using mixed-frequency methods

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Abstract

The delay in publication of Rwanda's gross domestic product figures has sparked the search for econometric tools to produce timely prediction of the current state of economic activity for timely policy making at the Ministry of Finance and Economic Planning. Nowcasting is a useful econometric tool that utilizes readily available information contained in high-frequency indicators to forecast the current (now) quarter GDP. In this paper we apply nowcasting techniques, namely bridge equations and a battery of mixed-frequency data sampling (MIDAS) regression models and mixed-frequency vector autoregressive (VAR)—a relatively recent methodology—to obtain current-quarter and next quarter nowcasts of Rwanda's real GDP growth by exploiting information contained in monthly macroeconomic and financial indicators. We then compare forecasting abilities (accuracy) of these techniques out of sample. Key to our findings is that a combination of bridge equation and unrestricted MIDAS forecasts gives best accuracy for current quarter nowcast of GDP.

Keywords: Bridge equations, GDP, MIDAS, nowcasting

JEL codes: E37, C53

1. Introduction

Gross domestic product (GDP) is a key statistic for macroeconomic decision-making and the primary and broadest measure of the economic activity. However, Rwanda's GDP is typically published by the National Institute of Statistics of Rwanda (NISR) on a quarterly basis and with three months of delay¹. During this delay, however, the Ministry of Finance and Economic Planning (MINECOFIN) needs information on the current stance of the economy for timely policy making. This need to know the state of the economy *now* has sparked interest in nowcasting techniques.

Nowcasting is defined as the current-quarter forecasting (Kurz-Kim, 2019) of GDP using readily available high frequency (typically monthly) macroeconomic and financial indicators, such as the index of industrial production, exports, money supply and credit to private sector. It is a useful tool for macroeconomic policy makers in the ministries of finance/economic planning and at central banks. The term is a contraction for now and forecasting and is defined as the prediction of the present, the very near future and the recent past (Giannone et al., 2008; Bańbura et al., 2013).

Nowcasting has been for a long time applied in meteorology and its application in economics is relatively recent. The methodological developments in this area has mainly sought to devise ways to exploit timely available information, deal with mixed frequency and ragged-edge data. The ragged-edge issue is due to the fact that data on different variables are released at different dates, which results in missing observations at the end of the sample for some variables (Allan et al., 2019; Ferrara et al., 2010).

To deal with these data irregularities, several methods have been proposed. The most convenient way to deal with differing frequencies when nowcasting quarterly GDP has been to use 'bridge' equations which bridge the gap between GDP series—which are published quarterly—and high frequency indicators (HFIs) —which are available monthly. For nowcasting quarterly GDP, monthly indicators are aggregated to obtain quarterly values which are then used on the right-hand side of the equation to predict GDP. This approach was first developed by Trehan (1989, 1992) and it has been popular in nowcasting and forecasting literature and among policy makers mainly

¹ The National Institute of Statistics of Rwanda (NISR), for example, published the 2019Q1 GDP in June, 2019; 2019Q2 GDP was released mid-September, 2019; 2019Q3 GDP figures were published in December, 2019 and the release of 2019Q4 GDP figures is expected in March, 2020. This schedule implies a lag of three months.

because its simplicity and transparency, implementation and the interpretation of results are straightforward (Kim and Swanson, 2015; Schumacher, 2016). This aggregation, however, is not without harm². By aggregating monthly to quarterly frequency, for example, useful information is lost.

To fully exploit the information contained in high frequency indicators, Ghysels et al (2004) introduced mixed-data sampling (MIDAS) regression³. These regressions use distributed lag polynomials to deal with mixed frequency data but also achieve parsimonious specification because of tight parameterization (Foroni et al., 2017). MIDAS regressions have been applied to forecast GDP using monthly macroeconomic and financial indicators (see, for example, Clements and Galvao, 2008, 2009). The MIDAS approach presents several advantages over bridge equations that help to improve predictions. First, while bridge equations are based on iterated forecasts, MIDAS is a direct multi-step nowcasting tool. Second, due to time aggregation, the weights of indicator observations in bridge equations are partly fixed, while MIDAS equations employ empirical weighting of high-frequency predictor observations with estimated functional lag polynomials (see, Schumacher, 2016).

Ghysels et al. (2016) extends single equation MIDAS models to a vector autoregressive (VAR) setting to exploit the information contained in dynamic and simultaneous relations in a system of macroeconomic variables. Mixed frequency VAR are observation-driven rather than shocks to latent process as it is the case for state-space⁴ representation that has been used to deal with mixed frequencies (see, for example, Mariano and Murasawa, 2003).

The present study builds on existing literature on mixed-frequency data modelling. The main objective is to contribute to the empirical literature on nowcasting in developing countries and to highlight the usefulness of mixed frequency techniques. Against this backdrop, we seek , to (i)

² Ghysels (2016) analyzes the costs of aggregation in terms of impulse response analysis. Consider a shock in the first month of, for example, the first quarter; this shock will obviously affect the second and the third quarter. In the end the total impact will not be the same for aggregate and non-aggregated HFIs.

³ Another approach to circumvent this aggregation issue consists of taking a large number of HFIs and extract a small number of factors that contain most of the information (high predictive power). These factors are then used in regression on GDP (see Giannone et al., 2008; Allan et al., 2019).

⁴ Low and high frequency variables are put together in a VAR model; low-frequency series are treated as high-frequency series with missing observations, then the Kalman filter and smoother is applied to estimate missing observations and generate the forecasts (see, Foroni and Marcellino, 2013).

illustrate bridge equations, MIDAS and mixed-frequency VAR modelling, (ii) predict the current (now) quarterly and one quarter ahead real GDP growth using available monthly business cycle indicators, (iii) evaluate the performance of bridge equations, MIDAS and mixed-frequency VAR in predicting current and one quarter ahead real GDP growth.

The rest of the paper is structured as follows. Section 2 discusses bridge equations, MIDAS and mixed-frequency VAR models applied to nowcasting. Section 3 describes the data, presents and discusses empirical results. Section 4 concludes the paper.

2. Mixed-frequency data nowcasting approaches

In the customary regression analysis, the left- and right-hand variables must be sampled at the same frequency. In this section we describe regression models that relate data sampled at different frequency. We outline two approaches, namely bridge equations (BE) and MIDAS. BE have been applied for quite some time while MIDAS models are relatively very recent and have attracted attention in nowcasting literature that led to the development of several competing models. For more details on mixed-frequency modelling, refer to Foroni and Marcellino (2013).

2.1. Bridge equations

A BE is a low-frequency linear regression that uses aggregated HFIs to predict GDP. For detailed description of BE application to nowcasting, see Baffigi et.al. (2004) and Kalisa and Uwase Nsanza (2018). The latter study applies BE to Rwanda's GDP. BE is specified as follows

$$y_t = \beta_0 + \lambda y_{t-1} + \beta(L)x_t^q + \varepsilon_t \quad (1)$$

where y is GDP, x indicates HFI and $\beta(L)$ is the lag polynomial.

In practice, first HFIs are transformed to the lower frequency. Here we distinguish flow from stock variables (see, Schumacher, 2016). For a stationary monthly flow variable x_t^m , the aggregate is given by

$$x_t^q = x_t^m + x_{t-1/3}^m + x_{t-2/3}^m \quad (2)$$

If x_t^m is a stock variable, its aggregate is simply $x_t^q = x_t^m$.

Then a bivariate regression is applied on quarterly HFI and GDP and each bivariate equation is saved. After that each equation (one for each HFI) is used to forecast GDP. Because HFIs are

published with delay, we run into a problem commonly known as called “ragged” or “jagged” edge (Wallis, 1986) whereby the time of the last observation differs for all or some HFIs. Univariate autoregressive models are therefore utilized to obtain predictions of missing monthly HFIs values.

2.2. Mixed-data sampling frequency model (MIDAS)

Mixed-frequency models are very recent and were introduced by Ghysels et al. (2006). Mixed-data sampling (MIDAS) regression allows the dependent and independent variables to be sampled at different frequencies. The regression consists of single horizon-specific equation that relates low-frequency variable to various lags of high-frequency indicator (Ghysels et al., 2007) through a lag polynomial operator to ensure parsimonious specifications (Andreou et al., 2010; Den Reijer and Johansson; 2019).

The basic MIDAS equation to forecast quarterly GDP growth rate Y after h quarters, $t + h$ using indicators that are sampled monthly, $m = 3$, is

$$Y_{t+h} = \beta_0 + \gamma y_t + \beta_1 B\left(L^{\frac{1}{m}}; \theta\right) x_{t+w}^{(m)} + \varepsilon_{t+h} \quad (3)$$

where w indicates the lead of the HFI (see Andreou et al., 2013); γy_t is the autoregressive term and $B\left(L^{\frac{1}{m}}; \theta\right) = \sum_{j=0}^{j^{max}} b(j, \theta) L^{\frac{j}{m}}$ is the lag polynomial.

The lag polynomials are used for weighting schemes, which are data-driven and serve to avoid parameter proliferation and reduce the likelihood of multicollinearity by only using a small number of parameters; the weight decreases as the variable is lagged further (Foroni and Marcellino. 2013) For the description of different weighting schemes that are available for MIDAS estimation, see Gysels et al. (2004). In this study we employ three variant of weighting schemes, namely Almon that uses polynomial distributed lag (PDL, polynomial degree 3) structure, Step Weighting (3 steps) that uses discrete steps to approximate the distributed lag pattern and Unrestricted lag order specification MIDAS (U-MIDAS) which allows to avoid parameter proliferation, especially in the presence of sampling frequencies that are very close, such as monthly vs. quarterly data.

MIDAS models are nonlinear and are thus typically estimated by nonlinear least squares (NLS) (Foroni et al., 2019). The unrestricted version of MIDAS known as U-MIDAS (Foroni et al. 2015),

however, can be estimated by simple OLS and allows to include several HFIs. U-MIDAS performs better when the difference in frequency is low; which makes it an ideal choice for the quarterly GDP/monthly HFIs case of this paper⁵.

2.3. Mixed-frequency VAR

To exploit the information contained in dynamic and simultaneous relations in a system of macroeconomic variables, the univariate case can be generalized to a multivariate setting based on VAR methodology. Following Ghysels (2016), we specify a finite order VAR which is very similar to a usual periodic VAR except that we construct a stacked vector as follows:

$$\begin{bmatrix} X_H(\tau_L, 1) \\ \cdot \\ \cdot \\ X_H(\tau_L, m) \\ X_L(\tau_L) \end{bmatrix} = A_0 + \sum_{j=1}^p A_j \begin{bmatrix} X_H(\tau_{L-j}, 1) \\ \cdot \\ \cdot \\ X_H(\tau_{L-j}, m) \\ X_L(\tau_{L-j}) \end{bmatrix} + \varepsilon(\tau_L) \quad (4)$$

where the vector is of $K_L + m * K_H$ dimensional VAR with p lags; the index H and L indicate high-frequency and low-frequency variables respectively. Since GDP is quarterly while high frequency indicators (HFIs) are published monthly with some delay, the above representation consists of stacking the months of January, February and March together with the first quarter GDP; April, May and June with the second quarter GDP and so on. For this approach, unlike for example bridge equations that require time aggregation, stationary monthly indicators are directly entered in a VAR and whenever there is an incoming indicator observation, the VAR is re-estimated

3. Data and empirical results

3.1. Dataset

In this exercise we seek to nowcast the current quarter GDP and forecast one quarter ahead. For example, to nowcast 2019Q4 GDP, we need monthly data for HFIs for the months of the same quarter, i.e. until December 2019. Table 1 provides a summary of the variables, their source,

⁵ There are several extensions of the MIDAS model that are not discussed in this paper, *inter alia*, the factor-MIDAS (see Marcellino and Schumacher, 2010), asymmetric MIDAS (see Ghysels et al., 2005), smooth transition MIDAS (see Galvão, 2013), MIDAS with step functions (see Forsberg and Ghysels, 2007).

publication delays and the starting date. For some HFIs data is available for September, October for others and the latest we have is November. We thus face the ragged-edge data issue highlighted earlier. As a solution, we use autoregressive (integrated) moving average (ARIMA) models to forecast missing HFIs values till December 2019. In practice, the results are revised when data is available.

Table 1. Data description

Indicator	Frequency	Delay	Source Institution	Starting date	Latest available
Real GDP	Quarterly	3 months	NISR	2006Q1	2019Q3
Agriculture exports, value	Monthly	One month	BNR	Jan-04	Oct-19
Agriculture exports, volume	Monthly	One month	BNR	Jan-04	Oct-19
Cement imports	Monthly	One month	BNR	Jan-06	Oct-19
Cement production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Consumer price index (CPI)	Monthly	10 days	NISR	Jan-04	Nov-19
CPI Food	Monthly	10 days	NISR	Jan-04	Nov-19
Credit to private sector	Monthly	One month	BNR	Jan-05	Nov-19
Electricity production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Exports, value	Monthly	One month	BNR	Jan-05	Oct-19
Exports, volume	Monthly	One month	BNR	Jan-05	Oct-19
Flour production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Index of Industrial Production (IIP)	Monthly	One month	NISR	Jan-11	Nov-19
Imports, value cif	Monthly	One month	BNR	Jan-05	Oct-19
Imports, volume	Monthly	One month	BNR	Jan-05	Oct-19
Imports, petroleum products	Monthly	One month	BNR	Jan-04	Oct-19
M3	Monthly	One month	BNR	Jan-05	Nov-19
Milk production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Modern beer production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Nominal exchange rate (RWF/USD)	Monthly	No delay	BNR	Jan-06	Nov-19
Nominal composite index of economic activity	Monthly	One month	BNR	Feb-12	Nov-19

Producer price index (PPI)	Monthly	One month	NISR	Dec-10	Nov-19
Real composite index of economic activity	Monthly	One month	BNR	Feb-12	Nov-19
Rice production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Soft drinks production	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Total Cement	Monthly	three months	NISR/BNR	Jan-06	Sept-19
Turnovers, industry	Monthly	One month	NISR	Jan-12	Oct-19
Turnovers, services	Monthly	One month	NISR	Jan-12	Oct-19
VAT receipts	Monthly	three months	RRA	Jul-04	Oct-19
Water	Monthly	three months	NISR/BNR	Jan-06	Sept-19

Notes. NISR: National Bank of Rwanda; NISR: National Institute of Statistics of Rwanda; RRA: Rwanda Revenue Authority

3.2. Nowcasting exercise

First, the predictive ability of each HFI is evaluated (see Table 1A in Appendix). Then ten HFIs with high predictive ability are selected. All the variables are transformed in first difference of natural logarithm. This transformation helps to overcome the issue of nonstationary, and nowcasts can be interpreted as growth rates.

Data is then divided into the estimation sample and the evaluations sample. For performance evaluation, the estimation sample is 2006Q1-2018Q3, and the evaluation sample is 2018Q4-2019Q3. We first estimate the model and use it to nowcast the current quarter and forecast one quarter ahead. This process is repeated, adding one observation each time, till the end of the evaluation sample. Results are saved, plotted against the actual values and model performance is assessed based on the root mean square errors (RMSE), computed as follows.

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (5)$$

where $\hat{y}_i$ and y_i are the forecast and the actual value respectively.

3.3. Comparison of nowcasting performance

To evaluate the performance of BE and MIDAS, we fit the models using a portion of data and evaluate their respective performance using observation out of the estimation sample. Figure 1 depicts current quarter nowcasts and one quarter ahead forecast against the actual quarter-quarter GDP growth. Visually, bridge equations perform better than the rest of the methods, followed by U-MIDAS. This is valid for both current quarter nowcast and one quarter ahead forecast. MIDAS that uses polynomial distributed lag performs relatively worse.

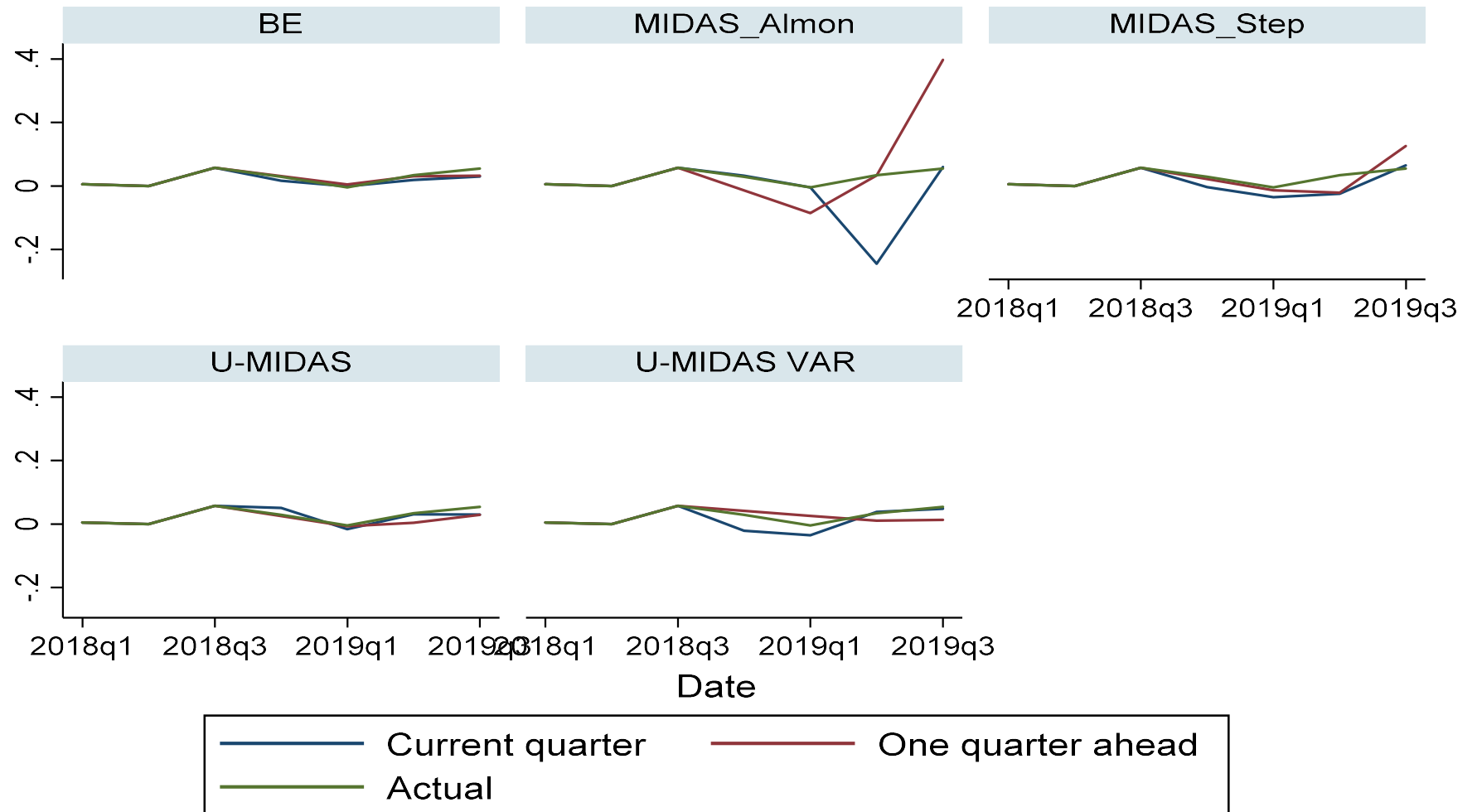


Figure 1. Out of sample forecasts

Table 2 ranks these methods based on RMSE. Not surprisingly, what one can observe looking at Figure 1 is corroborated in this table. Bridge equation outperforms other methods for both current quarter nowcasts and one quarter ahead forecast; the unrestricted MIDAS, comes second. A combination, based on average of forecasts, of BE and U-MIDAS performs best for current quarter nowcasting but does not outperform BE tout court for one quarter ahead forecast.

Table 2. Nowcast accuracy (ranked based on RMSE)

Current quarter		One quarter ahead	
BE&U-MIDAS	0.01356471	BE	0.014324868
BE	0.016074293	BE&U-MIDAS	0.017250277
U-MIDAS	0.017524265	U-MIDAS	0.023052658
U-MIDAS VAR	0.029426582	U-MIDAS VAR	0.032490849
MIDAS_Step	0.036947004	MIDAS_Step	0.052656529
MIDAS_Almon	0.139800882	MIDAS_Almon	0.20354688

Notes: BE&U-MIDAS indicates the combination (average) of BE and U-MIDAS forecasts.

4. Conclusion

In this paper we apply two classes of models, namely bridge equations and MIDAS to nowcasting Rwanda's quarterly GDP using monthly indicators of business cycle. These models were applied to produce current quarter nowcast and one quarter ahead forecast. For comparisons of model performance, forecasting accuracy was evaluated out of sample.

Our findings indicate that a combination of bridge equation and unrestricted MIDAS forecasts performs best for current quarter nowcasting. Overall, MIDAS class of models should be used as complementary to bridge equations.

References

- Allan, G., Koop, G., McIntyre, S., & Smith, P. (2019). Nowcasting using mixed frequency methods: an application to the Scottish economy. *Sankhya B*, 1-34.
- Andreou, E., Ghysels, E., & Kourtellos, A. (2010). Regression models with mixed sampling frequencies. *Journal of Econometrics*, 158(2), 246-261.
- Andreou, E., Ghysels, E., & Kourtellos, A. (2013). Should macroeconomic forecasters use daily financial data and how? *Journal of Business & Economic Statistics*, 31(2), 240-251.
- Baffigi, A., Golinelli, R., & Parigi, G. (2004). Bridge models to forecast the euro area GDP. *International Journal of forecasting*, 20(3), 447-460.
- Clements, M. P., and A. B. Galvao (2008). Macroeconomic forecasting with mixed-frequency data: Forecasting US output growth. *Journal of Business and Economic Statistics*, 26, 546-554.
- Clements, M. P., and A. B. Galvao (2009). Forecasting US output growth using leading indicators: An appraisal using MIDAS models. *Journal of Applied Econometrics*, 24(7), 1057-1217.
- Den Reijer, A., & Johansson, A. (2019). Nowcasting Swedish GDP with a large and unbalanced data set. *Empirical Economics*, 57(4), 1351-1373.
- Ferrara, L., Guegan, D., & Rakotomaroahy, P. (2010). GDP nowcasting with ragged-edge data: a semi-parametric modeling. *Journal of Forecasting*, 29(1-2), 186-199.
- Foroni, C., & Marcellino, M. (2013). A survey of econometric methods for mixed-frequency data. *Norge Bank Research Working Paper 06*, Norges Bank.
- Foroni, C., Marcellino, M., & Schumacher, C. (2015). Unrestricted mixed data sampling (MIDAS): MIDAS regressions with unrestricted lag polynomials. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 178(1), 57-82.
- Foroni, C., Marcellino, M., & Stevanovic, D. (2019). Mixed-frequency models with moving-average components. *Journal of Applied Econometrics*.
- Forsberg, L., & Ghysels, E. (2007). Why do absolute returns predict volatility so well? *Journal of Financial Econometrics*, 5(1), 31-67.

- Galvão, A. B. (2013). Changes in predictive ability with mixed frequency data. *International Journal of Forecasting*, 29(3), 395-410.
- Ghysels, E. (2016). Macroeconomics and the reality of mixed frequency data. *Journal of Econometrics*, 193(2), 294-314.
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2004). The MIDAS touch: Mixed data sampling regression models. Chapel Hill, N.C.: Mimeo.
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2006). Predicting volatility: Getting the most out of return data sampled at different frequencies. *Journal of Econometrics*, 131(1–2), 59–95.
- Ghysels, E., Sinko, A., & Valkanov, R. (2007). MIDAS regressions: Further results and new directions. *Econometric Reviews*, 26(1), 53-90.
- Giannone, D., Reichlin, L., & Small, D. (2008). Nowcasting: The real-time informational content of macroeconomic data. *Journal of Monetary Economics*, 55(4), 665-676.
- GiBańbura, M., Giannone, D., Modugno, M., & Reichlin, L. (2013). Now-casting and the real-time data flow. In *Handbook of economic forecasting* (Vol. 2, pp. 195-237).
- Habimana, O., Tabaro, D. and Kalisa, T. (2020). Nowcasting Rwanda's quarterly GDP using mixed-frequency methods. Paper presented at University of Rwanda, School of Economics Research Seminar Series in Applied Economics (Seminar no.1), February 11, 2020, Kigali, Rwanda.
- Kalisa, T. & Uwase Nsanza, J. (2018). Nowcasting GDP in Rwanda. *BNR Economic Review*, 12, 145-166.
- Kim, H. H., & Swanson, N. R. (2015). Methods for Pastcasting, Nowcasting and Forecasting Using Factor-MIDAS with an Application to Real-Time Korean GDP. mimeo, Rutgers University.[314].
- Koop, G., McIntyre, S., & Mitchell, J. (2019). UK regional nowcasting using a mixed frequency vector auto-regressive model with entropic tilting. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*.
- Kurz-Kim, J. R. (2019). A note on the predictive power of survey data in nowcasting euro area GDP. *Journal of Forecasting*, 38(6), 489-503.

- Marcellino, M., & Schumacher, C. (2010). Factor MIDAS for nowcasting and forecasting with ragged-edge data: A model comparison for German GDP. *Oxford Bulletin of Economics and Statistics*, 72(4), 518–550.
- Mariano, R., and Y. Murasawa (2003): A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics*, 18(4), 427-443.
- Schumacher, C. (2016). A comparison of MIDAS and bridge equations. *International Journal of Forecasting*, 32(2), 257-270.
- Trehan, B. 1989. Forecasting growth in current quarter real GNP. In: *Economic Review*. Federal Reserve Bank of San Francisco 1. 39–52.
- Trehan, B. 1992. Predicting contemporaneous output. In: *Economic Review*. Federal Reserve Bank of San Francisco 2. 3–11.
- Wallis, K. F. (1986). Forecasting with an econometric model: The ‘ragged edge’ problem. *Journal of Forecasting*, 5(1), 1-13.

Appendix

Table 1A. Predictive ability of HFIs

Evaluation statistics						
Forecast	RMSE	MAE	MAPE	SMAPE	Theil U1	Theil U2
REAL_GDPF_TURN_SERV	26.70863	22.47418	1.305175	1.305528	0.007848	0.555031
REAL_GDPF_REAL_CIEA	39.37814	36.24405	2.193578	2.165142	0.011454	0.860066
REAL_GDPF_NOM_CIEA	40.62479	36.78959	2.218253	2.188370	0.011813	0.865242
REAL_GDPF_EXP_VOL	41.37852	31.50703	1.873113	1.844216	0.012064	0.886055
REAL_GDPF_IMP_PETROL	45.15166	37.34448	2.205015	2.171206	0.013141	0.966190
REAL_GDPF_CPI_FOOD	46.59338	36.85684	2.182281	2.145241	0.013562	1.001147
REAL_GDPF_TOTAL_CEMENT	46.80104	42.89947	2.621926	2.581108	0.013598	1.011185
REAL_GDPF_RICE_PROD	50.68309	42.47430	2.497836	2.455865	0.014731	1.067898
REAL_GDPF_CEMENT_PROD	50.79757	41.28275	2.433646	2.391367	0.014768	1.077708
REAL_GDPF_WATER	51.84641	43.63783	2.601502	2.555550	0.015060	1.103448
REAL_GDPF_IMP_VOL	52.58907	44.83091	2.676417	2.628904	0.015269	1.123422
REAL_GDPF_FLOUR_PROD	53.30872	44.15986	2.613932	2.567256	0.015487	1.128232
REAL_GDPF_SOFT_DRINKS	53.53805	49.49580	2.986359	2.935763	0.015526	1.133203
REAL_GDPF_NER_USD	54.07812	49.12500	2.956278	2.906206	0.015691	1.118684
REAL_GDPF_CEMENT_IMP	54.31322	47.78930	2.885970	2.833728	0.015759	1.127169
REAL_GDPF_CPI	55.06486	44.78405	2.673727	2.621273	0.015988	1.177884
REAL_GDPF_VAT	55.71966	47.66234	2.848074	2.793975	0.016161	1.186502
REAL_GDPF_ELECTR_PROD	57.91822	50.66036	3.014503	2.958973	0.016798	1.232967
REAL_GDPF_PPI	58.42553	50.67662	3.026126	2.968648	0.016942	1.240508
REAL_GDPF_MILK	58.48485	51.16887	3.032836	2.975186	0.016951	1.246276

REAL_GDPF_MODERN_BEER	58.67514	50.64615	3.023789	2.964188	0.017005	1.245562
REAL_GDPF_IMP_CIF	59.53746	56.93835	3.460889	3.396754	0.017236	1.264201
REAL_GDPF_AGR_EXP_VAL	63.86275	55.01429	3.256715	3.187284	0.018478	1.353006
REAL_GDPF_EXP_VAL	64.06302	52.95371	3.119054	3.049651	0.018545	1.349503
REAL_GDPF_TURN_IND	66.52387	59.75550	3.573476	3.497116	0.019234	1.410823
REAL_GDPF_CREDIT_PRIV	66.97906	60.68136	3.680902	3.602057	0.019376	1.442274
REAL_GDPF_M3	68.33354	62.42941	3.776388	3.694958	0.019757	1.465727
REAL_GDPF_IIP	73.03926	64.62661	3.856159	3.763465	0.021065	1.569274
Simple mean	51.35058	45.30937	2.703235	2.657946	0.014907	1.090801
Trimmed mean	52.04186	45.85095	2.736240	2.689676	0.015105	1.106331
Simple median	52.42633	45.68618	2.722017	2.674953	0.015218	1.114528
Least-squares	NA	NA	NA	NA	NA	NA
Mean square error	45.23413	36.89802	2.173904	2.139479	0.013160	0.959428
MSE ranks	47.89153	40.78212	2.418386	2.379466	0.013919	1.016951

Notes: HFIs are ranked by predictive ability; highest predictive ability corresponds to lowest RMSE